

A discovery model

Given a database $\mathbf{r}$,

a language $\mathcal{L}$ for defining groups of the data, and

an interestingness criterion q for evaluating if a sentence $\phi \in \mathcal{L}$ is interesting,

Find $Th(\mathcal{L}, \mathbf{r}, q) = \{\phi \in \mathcal{L} \mid q(\mathbf{r}, \phi) = TRUE\}$.

A discovery model

In addition, we assume:

- A monotone specialization relation $\preceq$:
 $\preceq$ is a partial order on the sentences in $\mathcal{L}$.
If $\psi \preceq \phi$ and $q(\mathbf{r}, \phi) = TRUE$, then $q(\mathbf{r}, \psi) = TRUE$
- The sentences are represented as itemsets:
there exists a set R of items (attributes) and a function $f : \mathcal{L} \rightarrow \mathcal{P}(R)$ such that if $\psi \preceq \phi$ then $f(\psi) \subseteq f(\phi)$.

Given $\mathcal{L}$, $\mathbf{r}$, q and $\preceq$, compute $MTh(\mathcal{L}, \mathbf{r}, q, \preceq)$.

σ -frequent itemsets

Given a binary relation $\mathbf{r}$ and a support value $0 < \sigma < 1$, a **σ -frequent itemset** F is a itemset of columns (attributes) for which at least a σ fraction of the rows have value 1.

Relation					0.5-freq itemset			.25-freq itemset
	A	B	C	D	A	C	B	D
T_1	1	0	1	0	1	1	0	0
T_2	1	1	1	0	1	1	1	0
T_3	1	0	1	1	1	1	0	1
T_4	0	0	0	1	0	0	0	1
T_5	0	1	0	1	0	0	1	1
T_6	0	0	1	1	0	1	0	1
T_7	1	1	1	1	1	1	1	1
T_8	0	1	1	0	0	0	1	0

Let $Fr(\mathbf{r}, \sigma)$ be the set of σ -frequent itemsets
and $MFr(\mathbf{r}, \sigma)$ be the set of maximal σ -frequent itemsets.

The Problem:

Given a binary relation $\mathbf{r}$, and a support value $\sigma \in [0, 1]$, find all the maximal σ -frequent itemsets in $\mathbf{r}$, $MFr(\mathbf{r}, \sigma)$.

Computing Maximal σ -frequent itemsets

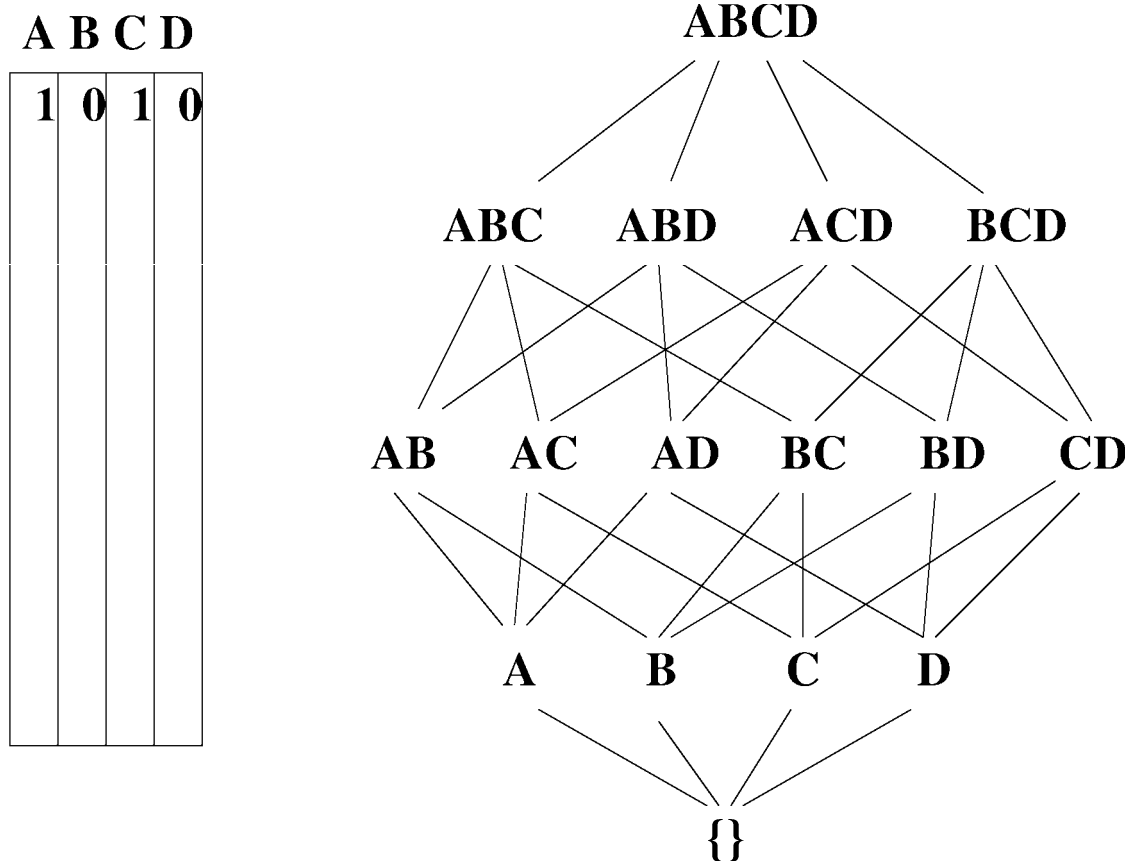
- Given a 0/1 relation r , and a threshold σ , the problem of finding the number of σ -frequent itemsets is #P-complete.
(Reduction from the MAX-2SAT problem.)
- Given a 0/1 relation r , and a threshold σ , the problem of deciding if there is a maximal σ -frequent itemset with at least T attributes is NP-complete.
(Reduction from the $K_{N,N}$ clique in bipartite graph problem.)

Complexity measures:

Output complexity.

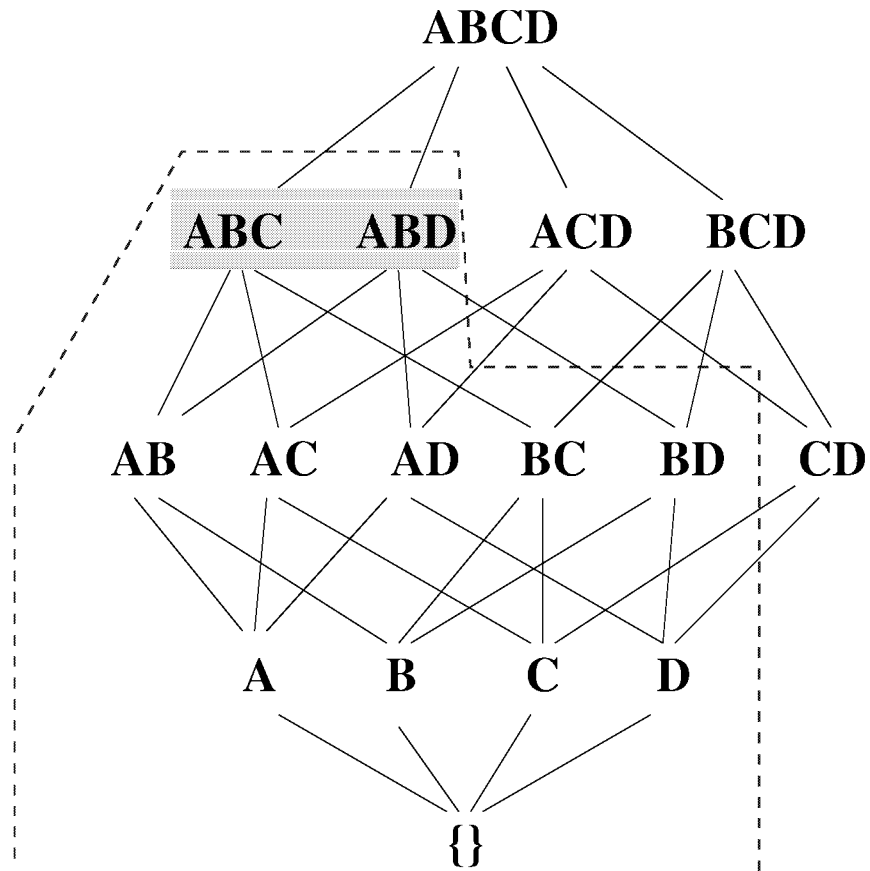
Number of ls-freq queries.

Computing Maximal σ -frequent itemsets



The *lattice* and the 0/1 relation matrix

A lower bound

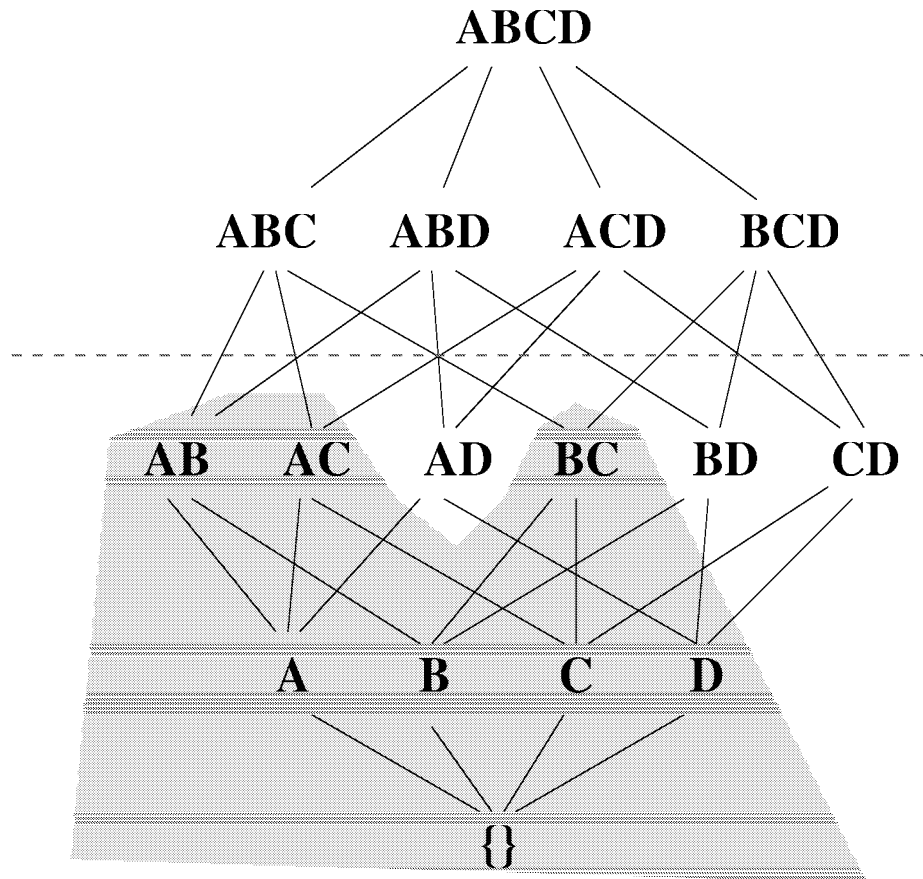


Let Bd^- be the minimal not σ -frequent itemsets, and

$$Bd = MFr(\mathbf{r}, \sigma) \cup Bd^-.$$

Computing $MF r(\mathbf{r}, \sigma)$ requires at least $|Bd|$ **Is-freq** queries.

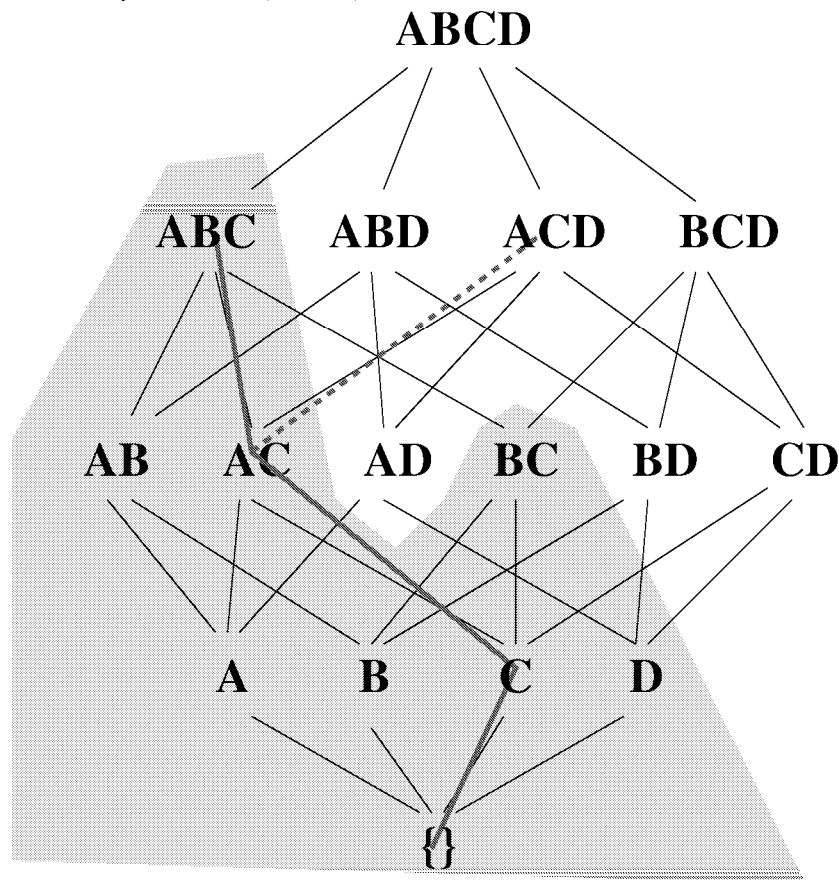
The *apriori* algorithm



1. Find the set F_1 of σ -frequent itemsets in level 1.
2. From F_i , compute the set C_{i+1} of possible σ -frequent itemsets in level $i + 1$
3. Find which itemsets from C_{i+1} are σ -frequent.

Find one maximal σ -freq itemset

Take a random ordering p of the attributes. Compute the lexicographically first, w.r.t. p , maximal σ -freq itemset (in $O(nm)$ time or m ls-freq queries)



For $p=(C,A,D,B)$ this is $\{A,B,C\}$.

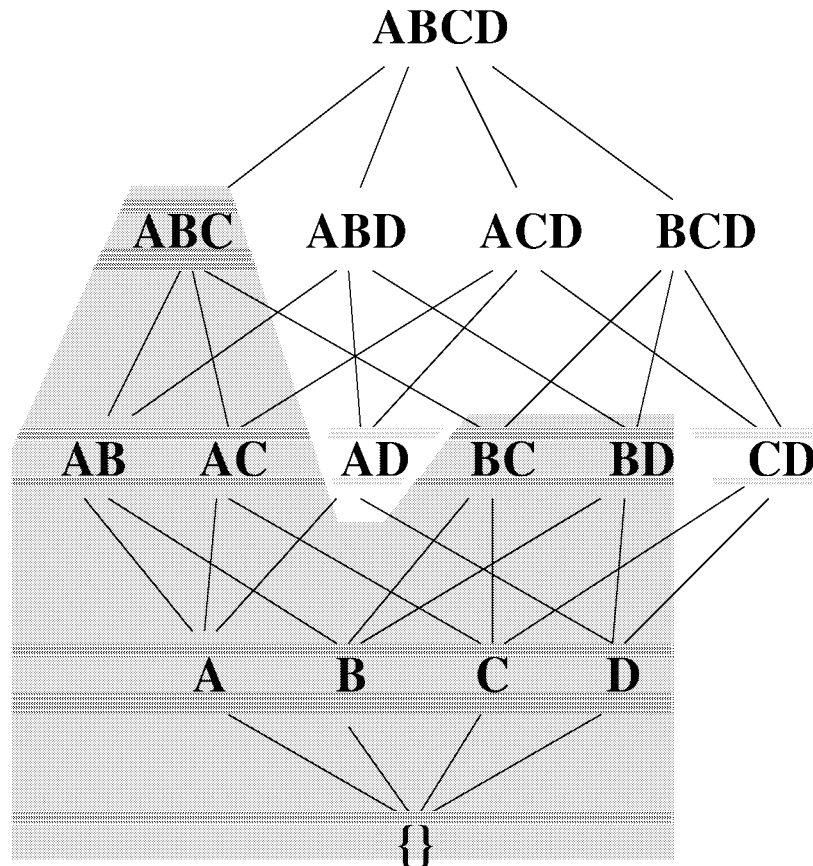
The randomized algorithm

- Algorithm Find_MFS:

Given a relation $\mathbf{r}$ with attributes $R_1, \dots, R_{|R|}$ and a threshold σ , find a maximal σ -frequent itemset.

1. Find a random permutation p of $(1, \dots, |R|)$:
2. Set $MFS = \{\}$.
3. For $i = 1$ to $|R|$:
 - (a) If $MFS \cup \{R_{p(i)}\}$ is a σ -frequent itemset, add $R_{p(i)}$ to MFS .
4. Return MFS

Computing $MF_{\sigma}(\mathbf{r}, \sigma)$



Maximal σ -frequent itemsets found so far:
 $\{A, B, C\}, \{B, D\}$.

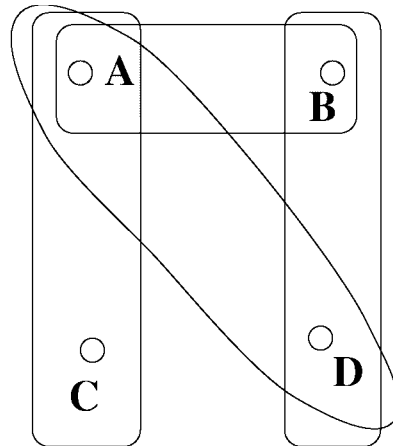
A new maximal σ -frequent itemset must include a itemset on the negative boundary ($\{A, D\}$ or $\{C, D\}$).

Transversals

A *hypergraph* is a 0-1 matrix with distinct rows. Each row is called a *hyperedge* and each column is called an *element*.

A *transversal* of a hypergraph is a set of elements such that every hyperedge contains at least one element from the set (every row has an 1 in at least one of the columns in the itemset.)

	A	B	C	D
E_1	1	0	1	0
E_2	1	0	0	1
E_3	1	1	0	0
E_4	0	1	1	0



Transversal: {A,B}

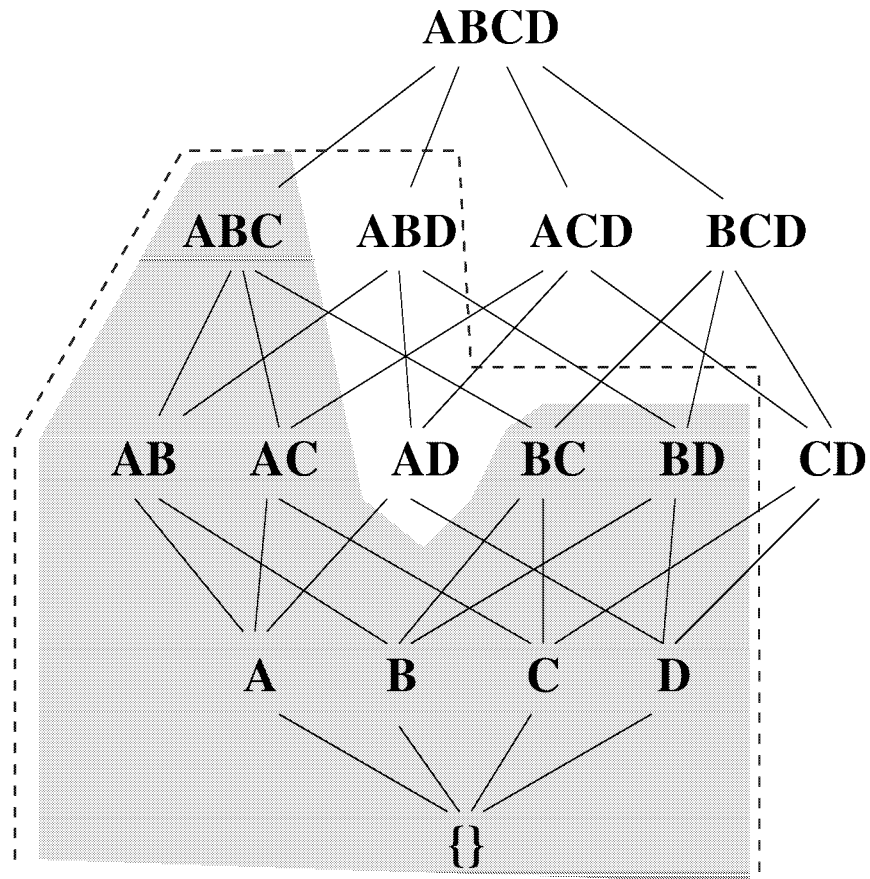
Transversals and maximal σ -frequent itemsets

Theorem

Given a set S of maximal frequent itemsets of a given matrix and support value, if there exists a maximal frequent itemset outside S , then the new frequent itemset *must contain a minimal transversal* of the hypergraph of complements of frequent itemsets in S .

Enumeration of all minimal transversals of a hypergraph with *polynomial delay* is a well known problem.

Transversals and maximal σ -frequent itemsets



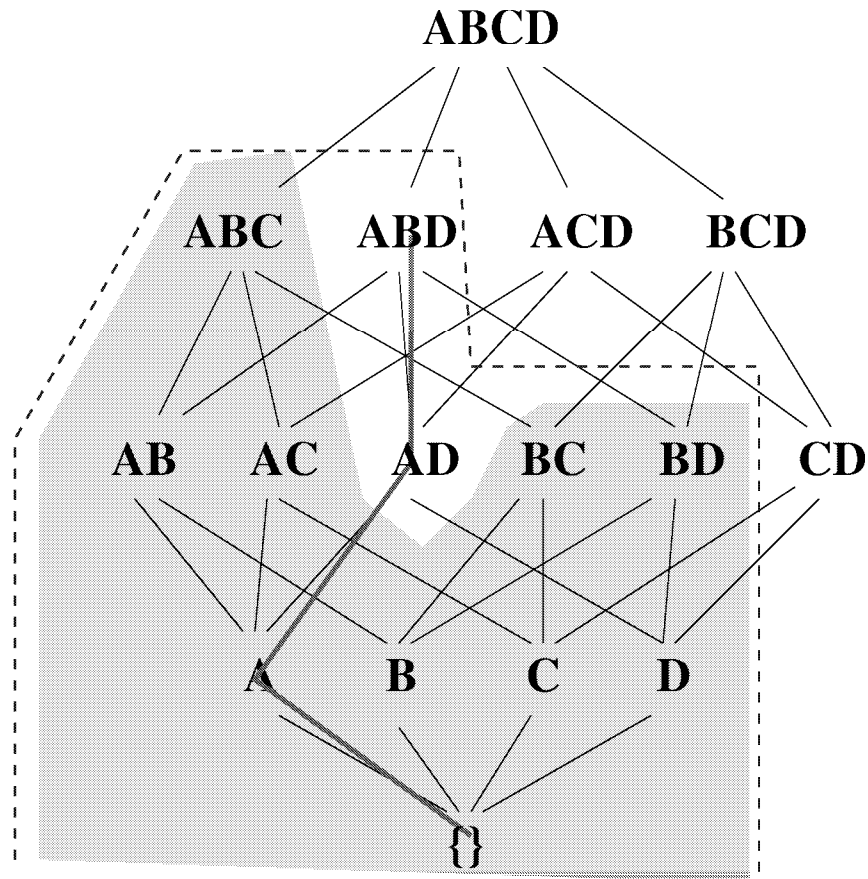
Each traversal either belongs to Bd^- or to $Fr(\mathbf{r}, \sigma)$.

Transversals and maximal σ -frequent itemsets

The following problems are equivalent:

- Computing maximal σ -frequent itemsets using only Is-freq queries.
- Computing hypergraph transversals using only Is-transversal queries.
Set $q(\mathbf{r}, \theta) = TRUE$ iff θ is not a transversal.
- Learning monotone boolean functions using only Membership queries.
Set $q(\mathbf{r}, \theta) = TRUE$ iff setting the variables in θ to 1 is a satisfying assignment.

Using the transversals



From the σ -frequent itemsets $\{A, B, C\}$ and $\{B, D\}$, find the hypergraph of the complements:

$$V = \{A, B, C, D\}, E = \{\{A, C\}, \{D\}\}$$

Its transversals are $\{A, D\}, \{C, D\}$.

$\{A, D\}$ is used as a “seed” for the ordering $\{A, D, B, C\}$.

Computing all minimal transversals

- Algorithm Find_Trans:
Given a hypergraph $G = (V, E)$,
 1. Compute the *lookahead matrix* for G .
 2. Run a backtracking algorithm using the lookahead matrix to output all transversals.

Experiments

Matrix Size	Transv. present	Time (sec)
58×12	97	0.5
97×12	58	1.5
420×22	1542	40.2

The Dualize and Advance algorithm:

Given a relation $\mathbf{r}$ with attributes $R_1, \dots, R_{|R|}$ and a threshold σ , compute $MFr(\mathbf{r}, \sigma)$.

1. Set $MFr(\mathbf{r}, \sigma) = \{\}$, $TR = \{\}$.
2. While new itemsets are added in $MFr(\mathbf{r}, \sigma)$ do:
 - (a) Find a transversal $TR = Tran(G_{R, \overline{MFr(\mathbf{r}, \sigma)}})$ that is not in $|Bd^-|$.
 - (b) Run algorithm Find_MFS on $\mathbf{r}$, TR .
 - (c) Add the result to $MFr(\mathbf{r}, \sigma)$
3. Return $MFr(\mathbf{r}, \sigma)$.

Theorem

The problem of computing $MFr(\mathbf{r}, \sigma)$ using only Is-freq queries can be solved with $O(m |MFr(\mathbf{r}, \sigma)| + |Bd^-|)$ Is-freq queries and $O(|MFr(\mathbf{r}, \sigma)|)$ calls to a subroutine for enumerating hypergraph transversals.

- In each iteration, a new maximal σ -frequent itemset is discovered.
- Algorithm Find_MFS makes $O(m)$ Is-freq queries.
- *Tran* is a transversal enumeration subroutine. Any set of $|Bd^-| + 1$ transversals must include a frequent itemset.
- We store the itemsets in Bd^- .

Using a recent result by Fredman and Khachiyan this gives an $O(|Bd|^{\log |Bd|})$ running time for the problem.

They give an incremental algorithm to compute the transversals of a hypergraph in $T(I, i) = (I + i)^{O(\log(I+i))}$.

The algorithm works for any $\mathcal{L}$ that can be represented as itemsets.

Experiments

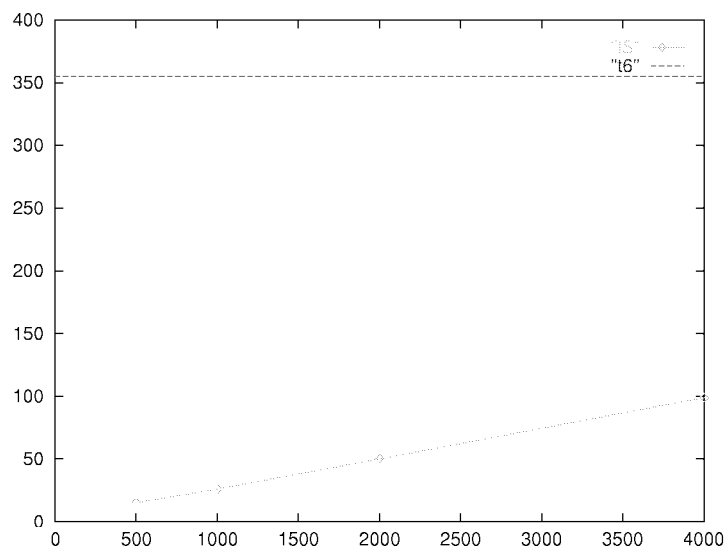
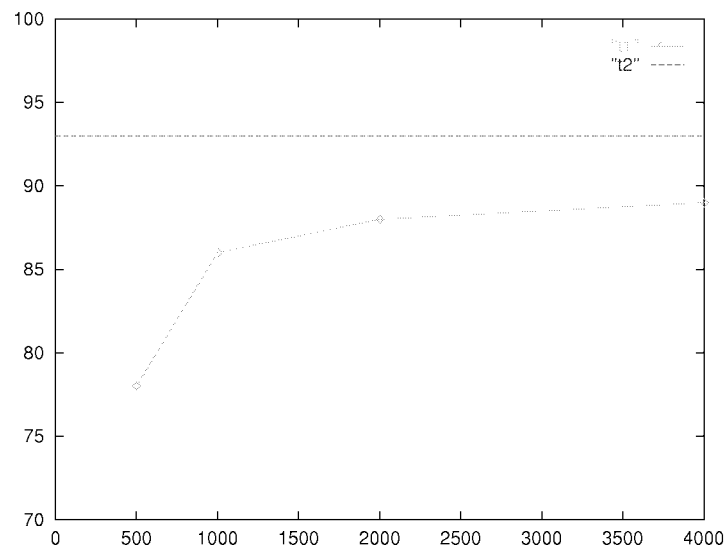
Matrix Size	σ (no. of rows)	Number of Runs	MFS found	Time (sec)
2670×20	100	500	78	15
2670×20	100	1000	86	26
2670×20	100	2000	88	50
2670×20	100	4000	89	99
2670×20	400	100	23	5
2836×129	100	500	178	56
2836×129	100	1000	244	108
2836×129	100	2000	283	208
2836×129	100	4000	303	409
2836×129	400	100	27	18

Experimental results for two data sets.

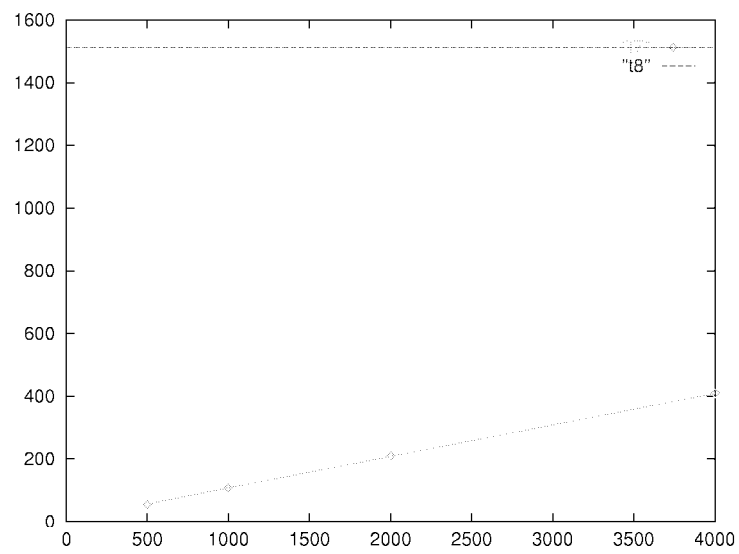
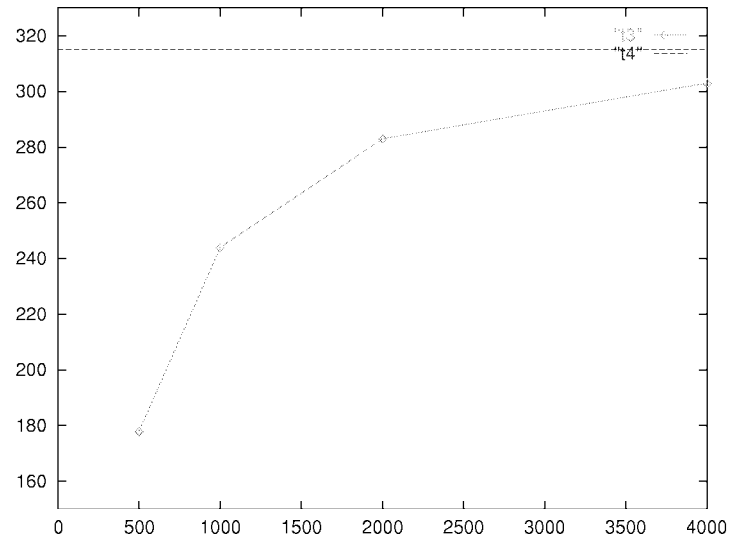
Matrix Size	s (no. of rows)	MFS found	Time (sec)
2670×20	100	93	355
2670×129	400	23	6
2836×20	100	315	1512
2836×129	400	27	10

The output of the apriori algorithm for the same data sets.

Experiments



Experiments



A generalized result

Given a database $\mathbf{r}$,
a language $\mathcal{L}$,
and a frequency criterion q .

Let $Th(\mathcal{L}, \mathbf{r}, q) = \{\phi \in \mathcal{L} \mid q(\mathbf{r}, \phi) = true\}$

Assume that there is a partial order $\preceq$, such that q is monotone
w.r.t. $\preceq$
(for all $\psi, \theta \in \mathcal{L}$, if $\psi \preceq \theta$ and $q(\mathbf{r}, \psi)$ then $q(\mathbf{r}, \theta)$)

Theorem

The Dualize and Advance algorithm computes all the most specific sentences in $Th(\mathcal{L}, \mathbf{r}, q)$.

Summary

- The Dualize and Advance algorithm is efficient if the interesting itemsets are large.
- Transversals are important for frequent itemsets.
- The results hold for the more general case.

Problems for future work

- Strategies to handle sparse matrices efficiently.
- Lower bounds for a more realistic model.
- Use similar techniques for more complicated problems (sequential patterns).